Chapter 10

Troubleshooting Linux

Episode 10.01

Episode title: **Troubleshooting Linux Systems**

Objective: 5.1

Troubleshooting Overview

Role of troubleshooting in Linux administration

Monitoring concepts and visibility

Troubleshooting Methodology & Monitoring Principles

Structured troubleshooting approach

Baselines and normal behavior

Reactive vs proactive monitoring

Service Monitoring Metrics (SLA, SLI, SLO)

SLA defines service availability commitment

SLI measures service performance

SLO sets target performance goals

Used to evaluate service health

System logs, Journals, and Service Status

Role of logs in troubleshooting

systemd-journald and journaldctl

Service status and dependencies

Monitoring Data Acquisition Methods

SNMP for device monitoring

SNMP traps and MIBs

Agent and agentless monitoring

Webhooks and health checks

Log aggregation

Resource Monitoring: CPU, Memory, and Disk

CPU and process monitoring

Memory usage and pressure

Disk utilization and saturation

Monitoring Configuration Elements

Thresholds define limits

Alerts trigger notifications

Events record state changes

Notifications inform responders

Logging supports investigation

Monitoring & Initial Troubleshooting (DEMO)

Reactive vs proactive monitoring

Baselines and normal system behavior

Logs, journals, and service status

Correlating symptoms with metrics

Troubleshooting Summary

Start with monitoring and visibility

Choose tools based on symptoms

Episode 10.02

Episode title: **Troubleshooting Linux Systems Demo**

Objective: 5.1

Episode 10.03

Episode title: **Troubleshooting Hardware**

Objective: 5.2

Troubleshooting Hardware Overview

Role of hardware troubleshooting in Linux

Visibility between hardware and kernel

Common Linux OS Failure Scenarios

- Kernel panic or corruption
- GRUB misconfiguration
- Package dependency failures
- PATH misconfiguration
- Killed processes
- Quota issues
- Segmentation faults

Additional Hardware & OS Failure Symptoms

Server not turning on

Server inaccessible

Missing or disabled drivers

Device failure indicators

Filesystem not writable

Hardware Detection & Enumeration

Device discovery at boot

PCI, USB, and block devices

Kernel and udev role

Kernel Messages & Hardware Errors

`dmesg` output

Kernel ring buffer

Common hardware error indicators

CPU, Memory, & Peripheral Issues

CPU-related symptoms

Memory availability and errors

Peripheral device failures

Hardware Troubleshooting Workflow (DEMO)

- Hardware vs software fault isolation
- Kernel detection and error reporting
- Firmware and device visibility
- CPU, memory, disk, and peripheral diagnostics
- Evidence-based troubleshooting workflow

Episode 10.04

Episode title: **Troubleshooting Hardware Demo**

Objective: 5.2

Episode 10.05

Episode title: **Troubleshooting Storage**

Objective: 5.2

Troubleshooting Storage Overview

Storage troubleshooting scope

Storage layers in Linux

Block Devices, Partitions, & Device Visibility

Block device identification

Partitions and logical volumes

Missing or failed disks

Filesystems, Mount Points, & Capacity Issues

Mounted vs unmounted

Disk space and inode exhaustion

Read-only filesystem states

Persistent Mounts, fstab, & Boot Failures

- /etc/fstab misconfiguration
- UUID usage and stability
- Boot and mount failures

Troubleshooting Storage Workflow (DEMO)

Storage stack visibility (disk → partition → filesystem → mount)

Capacity vs availability vs access issues

Filesystem and mount troubleshooting

Configuration vs hardware vs usage problems

Evidence-based troubleshooting workflow

Episode 10.06

Episode title: **Troubleshooting Networking**

Objective: 5.3

Troubleshooting Performance Overview

Networking troubleshooting scope

Network stack visibility in Linux

Network Interfaces, Links, & IP Addressing

Interface state and link status

IP addressing and subnetting

Common interface failures

Common Network Failure Scenarios

- Misconfigured firewall rules
- DHCP address failures
- DNS resolution problems
- MTU mismatch
- IP conflicts
- Dual-stack IPv4 and IPv6 issues
- Cannot ping server

Advanced Network Connectivity Issues

- Bonding misconfiguration
- MAC address spoofing
- Gateway and routing issues
- Link down or negotiation failures
- Server unreachable

Routing, Gateways, & Network Reachabililty

Routing tables

Default gateways

Path verification

Troubleshooting Networking Workflow (DEMO)

Network stack troubleshooting (interface → IP → routing → DNS → service)

Connectivity vs name resolution issues

Local vs remote network failures

Evidence-based diagnosis

Episode 10.07

Episode title: **Troubleshooting Performance**

Objective: 5.4

Troubleshooting Performance Overview

Performance troubleshooting scope

Identifying bottlenecks in Linux

Performance Symptoms and Indicators

High CPU usage or load average

Excessive context switching

Memory pressure or swapping

High I/O wait time

Slow startup and response times

Blocked or unresponsive processes

Network and Storage Performance Issues

Packet drops and jitter

Random disconnects and timeouts

High latency

Low throughput

Slow remote storage access

High disk latency

Baselines, Errors, and Resource Limits

Exceeding performance baselines

Hardware error indicators

Sluggish terminal behavior

CPU bottlenecks

CPU Load, Processes, & Scheduling

CPU utilization vs load average

Process states and priorities

Identifying CPU bottlenecks

Memory Pressure, Swap, and Paging

Memory utilization concepts

Swap activity and impact

Out-of-memory conditions

Disk I/O and Filesystem Performance

I/O wait and latency

Disk throughput bottlenecks

Identifying slow storage

Troubleshooting Performance Workflow (DEMO)

Performance symptoms vs root causes

CPU, memory, disk I/O, and load analysis

Interpreting system metrics

Correlating performance data across subsystems

Episode 10.07

Episode title: **Troubleshooting Security**

Objective: 5.5

Troubleshooting Security Overview

Scope of security troubleshooting

Linux security layers and visibility

Permissions, Ownership, & Access Control

File permissions and ownership

Common permission-related failures

Diagnostic commands

Authentication, sudo, & Account Restrictions

Authentication failures

`sudo` configuration issues

Account and login restrictions

SELinux, AppArmor, & Firewall Enforcement

Mandatory access controls

SELinux/AppArmor enforcement modes

Firewall-related security blocks

Common Linux Security Failure Scenarios

- SELinux policy or context errors
- Permission and ACL issues
- Account access failures
- Certificate problems
- Exposed or misconfigured services
- Insecure protocols or ciphers
- Cipher negotiation failures

Additional Security Troubleshooting Scenarios

- Unpatched vulnerable systems
- Misconfigured package repositories
- Remote access failures
- Firewall blocking legitimate traffic

Troubleshooting Security Workflow (DEMO)

- Authentication vs authorization vs accounting failures
- Permissions and ownership issues
- Service access vs network access
- Firewall, sudo, and login troubleshooting
- Evidence-based security diagnosis

Episode 10.08

Episode title: **Documentation**

Objective: 5.1

Documentation Overview

Role of documentation in Linux administration

Operational continuity and knowledge transfer

Relationship to troubleshooting and compliance

Documentation Exam Objectives

Recognize types of documentation

Where to find system-level documentation on Linux hosts

How to maintain and verify documentation quality

Documentation Types and Purposes

- System documentation
- Runbooks and playbooks
- SOPs and policies
- Troubleshooting guides and knowledge bases
- Configuration and API documentation
- Built-in Linux documentation

Finding System Documentation

- `man` pages
- `info` pages
- `/usr/share/doc`
- `systemd` unit documentation
- `/etc` and `/var` comments
- Vendor and upstream documentation

Maintaining Effective Documentation

- Clarity and structure
- Plain-text formats (Markdown)
- Exact commands and expected output
- Version control and change history
- Review and maintenance processes

Documentation Summary

Documentation is an operational control

Prefer versioned, verified documentation

Use built-in Linux docs first

Build, Version, and Audit a Runbook (DEMO)

- Runbooks vs procedures vs notes
- Documentation as an operational control
- Version control for documentation
- Auditability and verification
- Repeatable, testable operational steps